

Bridge to IGCSE — Science & Physics Skills

REVISION SHEET · WHAT TO KNOW

For Grade 8 / Year 8 scientists making the jump from Checkpoint to IGCSE.

Learn these core skills before you start IGCSE. Most are useful across **all three sciences**; the last section is **Physics-specific**. Tick them off as you grow confident.

1 · Maths fluency

- **Rearranging equations:** do the same to both sides. From $v = s/t \rightarrow$ multiply by $t \rightarrow s = vt \rightarrow$ divide by $v \rightarrow t = s/v$.
- **Unit conversions:** $1 \text{ km} = 1000 \text{ m}$ · $1 \text{ m} = 100 \text{ cm}$ · $1 \text{ kg} = 1000 \text{ g}$ · $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$ · $1 \text{ min} = 60 \text{ s}$. Convert to SI **before** calculating.
- **Standard form:** $a \times 10^n$ with $1 \leq a < 10$. $4500 = 4.5 \times 10^3$, $0.00045 = 4.5 \times 10^{-4}$.
- **Significant figures:** count from the first non-zero digit; quote answers to 2–3 s.f., matching the data.
- **Proportionality:** directly proportional $\rightarrow$ straight line through the origin (double one, double the other). Inversely proportional $\rightarrow$ curve (double one, halve the other).
- **Percentages:** $\% = (\text{part} \div \text{whole}) \times 100$. $\% \text{ change} = (\text{change} \div \text{original}) \times 100$.
- **Estimate:** always check the answer is a sensible size (right order of magnitude).

2 · Command words

The command word tells you exactly what kind of answer earns the marks.

State / Define	Give a short fact or the meaning of a term. No explanation needed.
Describe	Say WHAT happens or what you observe — the facts, in order.
Explain	Say WHY it happens — give the reason or mechanism.
Calculate	Work out a number. Show working and give the unit.
Determine	Work out a value using given data, a graph or a measurement.
Compare	Give similarities AND differences — link the two things.
Suggest	Apply what you know to a new or unfamiliar situation.

3 · Graphs & handling data

- **Axes:** the variable you change (independent) goes on the **x-axis**; the one you measure (dependent) on the **y-axis**.
- **Scales:** choose easy intervals that fill most of the grid; label each axis with **quantity AND unit**.
- **Best fit:** draw a smooth line or curve balancing the points; don't just join the dots.
- **Gradient** = $\Delta y \div \Delta x$. Gradient of distance–time = speed; gradient of speed–time = acceleration.
- **Area** under a speed–time graph = distance travelled.

4 · Measuring & practical work

- **Parallax:** read a scale with your eye level with it to avoid a parallax error.
- **Meniscus:** read a measuring cylinder at the **bottom** of the curve, at eye level.
- **Repeat & mean:** take repeats, ignore any anomaly, then mean = sum $\div$ number of readings.
- **Accuracy** = close to the true value. **Precision** = repeat readings close together / fine resolution.
- **Errors:** random (scatter — reduce by repeating) vs systematic (same direction each time, e.g. a zero error).

5 · Planning an investigation

- **Variables:** independent = the one you change; dependent = the one you measure; control = kept the same to make it a **fair test**.
- **Prediction/hypothesis:** a testable statement of what you expect to happen and why.
- **Tables:** headed columns with units; independent variable in the left column.
- **Result vs conclusion:** the result is the data; the conclusion is what the data tells you about your prediction.
- **Evaluate:** note limitations, sources of error, and improvements.

6 · Communicating like a scientist

- **Show your working** — method marks survive an arithmetic slip.
- Always give the **correct unit** and a sensible number of significant figures.
- Use correct terms; explain with “**because... so...**” cause-and-effect sentences.

7 · Physics foundations

- A physical quantity = **a number AND a unit** (5 m, not just 5).
- **SI base units:** metre (m), kilogram (kg), second (s), ampere (A), kelvin (K).
- **Scalar** = size only (distance, speed, mass, time, energy). **Vector** = size + direction (displacement, velocity, force, acceleration).
- **Density:** $\rho = m / V$ — units kg/m³ or g/cm³.
- **Speed:** $v = s / t$ — units m/s.
- **Energy** is stored (kinetic, gravitational, chemical, elastic, thermal) and transferred (mechanically, electrically, by heating, by radiation). It is **conserved** — the total never changes.

You've got everything you need for this. Wishing all my Year 8 scientists the very best for IGCSE. — Semra SG